Project Report

Customer Segmentation Using RFM Analysis and Unsupervised Machine Learning

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Domain: Consumer Segmentation and Consumer Behavior

Dataset: UCI Online Retail via Kaggle | Platform: Google Colab | Language: Python 3

1. Executive Summary

This report presents a comprehensive customer segmentation analysis of a UK-based e-commerce retailer using the RFM (Recency, Frequency, Monetary) framework combined with three unsupervised machine learning algorithms: K-Means, Hierarchical Clustering, and DBSCAN. The analysis was conducted on 392,692 cleaned transactions spanning December 2010 to December 2011, covering 4,338 unique customers across 37 countries.

K-Means clustering with k=4 was identified as the optimal model based on Silhouette Score (0.334), Davies-Bouldin Score (1.016), and Calinski-Harabasz Score (3286.68). Four behaviorally distinct segments were identified: Champions, Loyal Customers, New Customers, and Lost Customers. The most critical finding is that Champions, representing 16.76% of customers, generate 65.79% of total revenue, consistent with the Pareto principle in consumer behavior literature.

2. Introduction

2.1 Background and Motivation

Customer segmentation is one of the most studied problems in marketing research. The ability to identify homogeneous subgroups within a heterogeneous customer base enables firms to allocate marketing resources efficiently, personalize communications, and maximize customer lifetime value. Traditional segmentation approaches rely on demographic or psychographic variables requiring primary data collection. Behavioral segmentation using transactional data offers a scalable and objective alternative directly tied to observed purchasing behavior.

The RFM model, first formalized in direct marketing literature, operationalizes customer value across three dimensions: how recently a customer purchased, how frequently they purchase, and how much monetary value they generate. This project extends the classical RFM scoring approach by applying machine learning clustering algorithms to derive data-driven segments rather than relying on arbitrary score thresholds.

2.2 Research Questions

- How can raw transactional data be transformed into behavioral customer profiles using the RFM framework?
- Which clustering algorithm produces the most statistically valid and business-interpretable customer segments?
- What are the managerial implications of data-driven segmentation for targeted marketing strategy?

2.3 Dataset Description

The dataset was sourced from the UCI Machine Learning Repository via Kaggle. It contains transactional records from a UK-based online retailer specializing in gift items for the period December 2010 to December 2011. The original dataset contained 541,909 records across 8 fields: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country.

3. Data Preprocessing and Cleaning

3.1 Data Quality Assessment

A thorough data quality audit was conducted on the raw dataset. The following issues were identified and resolved:

Issue	Count	Percentage of Total
Missing CustomerID	135,080	24.93%
Duplicate rows	5,225	0.96%
Cancelled transactions (InvoiceNo prefix C)	8,872	1.71%
Invalid Quantity or UnitPrice (zero or negative)	40	0.01%

3.2 Sequential Cleaning Pipeline

Step	Action	Rows Before	Rows After	Removed
1	Remove missing CustomerID	541,909	406,829	135,080
2	Remove duplicate rows	406,829	401,604	5,225
3	Remove cancelled transactions	401,604	392,732	8,872
4	Remove invalid Quantity/UnitPrice	392,732	392,692	40

The final cleaned dataset retained 392,692 rows, representing 72.46% of the original data. It covers 4,338 unique customers, 18,532 unique invoices, 3,665 unique products, and 37 countries.

4. Exploratory Data Analysis

4.1 Revenue Overview

Total revenue across the period was GBP 8,887,208.89. The average order value was GBP 479.56 and the average revenue per customer was GBP 2,048.69. The maximum single transaction value of GBP 168,469.60 indicates the presence of high-value wholesale or bulk buyers.

4.2 Temporal Patterns

Monthly transaction volume showed a clear upward trend in Q4 2011, peaking in November with 2,657 transactions and revenue of approximately GBP 1.14 million. The apparent decline in December 2011 is a data artifact as the dataset covers only the first nine days of that month.

4.3 Geographic Distribution

The United Kingdom accounted for approximately GBP 7.3 million in revenue, reflecting the retailer's primary market. International markets including the Netherlands, EIRE, Germany, and France contributed smaller but meaningful revenue proportions.

4.4 Customer Revenue Distribution

The distribution of revenue per customer is heavily right-skewed with a long tail of high-spending customers. This pattern motivates the use of clustering to identify and prioritize high-value segments for targeted marketing investment.

5. RFM Feature Engineering

5.1 RFM Metric Computation

RFM metrics were computed at the customer level using December 10, 2011 as the reference date, one day after the last transaction in the dataset. This ensures all customers carry a positive recency value.

Metric	Mean	Median	Min	Max	Std Dev
Recency (days)	92.54	51.00	1	374	100.01
Frequency (orders)	4.27	2.00	1	209	7.70
Monetary (GBP)	2,048.69	668.57	3.75	280,206.02	8,985.23

5.2 Log Transformation and Scaling

All three RFM metrics exhibited strong positive skewness: Recency (1.25), Frequency (12.07), and Monetary (19.34). Log1p transformation reduced skewness substantially to -0.38, 1.21, and 0.40 respectively. RobustScaler was applied for standardization, chosen over StandardScaler for its resistance to extreme outliers prevalent in retail transactional data.

6. Clustering Analysis

6.1 K-Means Clustering

K-Means was applied using k-means++ initialization with 10 random restarts. The elbow method and silhouette scoring both supported k=4 as the optimal cluster count. While k=2 produced the highest silhouette score (0.430), it offers insufficient business granularity. At k=4, the silhouette score of 0.334 balances statistical validity with interpretability.

6.2 Hierarchical Clustering

Agglomerative Hierarchical Clustering with Ward linkage was applied at k=4 for comparability. A dendrogram on a 200-customer sample confirmed the 3 to 4 cluster structure. The algorithm achieved a Silhouette Score of 0.228, below K-Means, indicating less compact cluster structure.

6.3 DBSCAN

DBSCAN was configured with epsilon=0.25 derived from the k-distance graph elbow and min_samples=5. It identified three main clusters and flagged 130 customers (3.00%) as noise points. Two micro-clusters of 4 and 5 customers were identified but are not actionable. DBSCAN underperformed K-Means on all evaluation metrics.

6.4 Algorithm Comparison

Algorithm	Silhouette Score	Davies-Bouldin	Calinski-Harabasz
K-Means (k=4)	0.3340	1.0156	3286.68
Hierarchical (k=4)	0.2275	1.2293	2534.61
DBSCAN (eps=0.25)	0.1034	1.2507	1037.86

K-Means achieved the best performance across all three metrics and was selected as the final segmentation model.

7. Segmentation Results and Business Interpretation

7.1 Segment Profiles

Segment	Customers	Avg Recency	Avg Frequency	Avg Monetary	Revenue Share
Champions	727 (16.76%)	13 days	14 orders	GBP 8,042	65.79%
Loyal Customers	1,220 (28.12%)	76 days	4 orders	GBP 1,714	23.52%
New Customers	839 (19.34%)	18 days	2 orders	GBP 560	5.29%
Lost Customers	1,552 (35.78%)	183 days	1 order	GBP 309	5.40%

7.2 Segment Descriptions and Strategy

Champions

Champions purchased an average of 13 days ago, place 14 orders on average, and generate GBP 8,042 per customer. Despite being 16.76% of the base, they generate 65.79% of revenue. Recommended strategy: retention through exclusive loyalty programs, early product access, and personalized outreach.

Loyal Customers

Loyal Customers contribute 23.52% of revenue with moderate recency (76 days) and frequency (4 orders). This segment presents the highest growth opportunity. Recommended strategy: upselling, cross-selling, and frequency incentives to accelerate progression toward Champion status.

New Customers

New Customers purchased recently (18 days) but show low frequency (2 orders) and moderate spend (GBP 560), indicating an undeveloped purchase habit. Recommended strategy: onboarding sequences and first repeat purchase incentives delivered within 30 days of acquisition.

Lost Customers

Lost Customers last purchased 183 days ago on average, have minimal frequency (1 order), and contribute only 5.40% of revenue despite comprising 35.78% of the base. Recommended strategy: selective win-back campaigns with strong incentives, prioritizing those with higher historical spend.

8. Critical Analysis

8.1 Strengths

- The preprocessing pipeline is thorough and reproducible. Each cleaning step is individually quantified, enabling readers to audit and replicate the process.
- RobustScaler is correctly chosen over StandardScaler given the confirmed presence of extreme outliers across all three RFM dimensions.
- Three complementary evaluation metrics are used in combination, reducing the risk of selecting a suboptimal model based on a single criterion.
- Log transformation is applied before scaling, which is the correct sequence for distance-based algorithms on skewed data.
- Business interpretation is grounded in the statistical results, connecting clustering outputs to actionable marketing strategy.

8.2 Limitations and Areas for Improvement

- CustomerID 12346 is classified as Loyal Customer despite a recency of 326 days and a single transaction of GBP 77,183. This edge case illustrates how a single large transaction distorts segment assignment and warrants a separate outlier treatment rule.
- The notebook structure declared in the opening section lists 13 sections while the implementation delivers a slightly different structure, which could create confusion for readers or reviewers.
- DBSCAN epsilon was selected visually from the k-distance plot. A more rigorous approach would test a range of epsilon values and report evaluation scores for each, allowing principled selection.
- The analysis covers a single year of data. Genuinely lost customers cannot be reliably distinguished from seasonal buyers without at least two to three years of transaction history.
- No cluster stability analysis is performed. Bootstrap resampling or repeated K-Means runs with different seeds would strengthen confidence in the identified segment structure.
- The UK versus international split is not analyzed. Given that the UK generates approximately 82% of revenue, a country-stratified segmentation could reveal whether segment compositions and marketing priorities differ across geographies.

9. Conclusion and Research Implications

This project demonstrates that combining the RFM behavioral framework with unsupervised machine learning produces statistically valid and strategically actionable customer segments from raw e-commerce transactional data. K-Means with k=4 was selected as the optimal model and yielded four distinct segments with clear behavioral profiles.

The finding that 16.76% of customers generate 65.79% of total revenue has direct implications for marketing resource allocation. Disproportionate investment in Champion retention and Loyal Customer conversion is expected to yield substantially higher returns than broad-based acquisition campaigns targeting the Lost Customer segment.

From a research perspective, this project contributes to the literature on data-driven consumer segmentation, illustrating the practical advantages of K-Means over density-based and hierarchical approaches for retail data characterized by extreme skewness and outlier prevalence. Future extensions could incorporate product category variables, customer demographics, or sequential purchase modeling to construct richer multi-dimensional behavioral segments.

10. References

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